NATURAL GAS ANALYST

US-Iran Agreement Paves Way For European Storage Refill

Samantha Dart
+1(212)357-9428 |
samantha.dart@gs.com
Goldman Sachs & Co. LLC

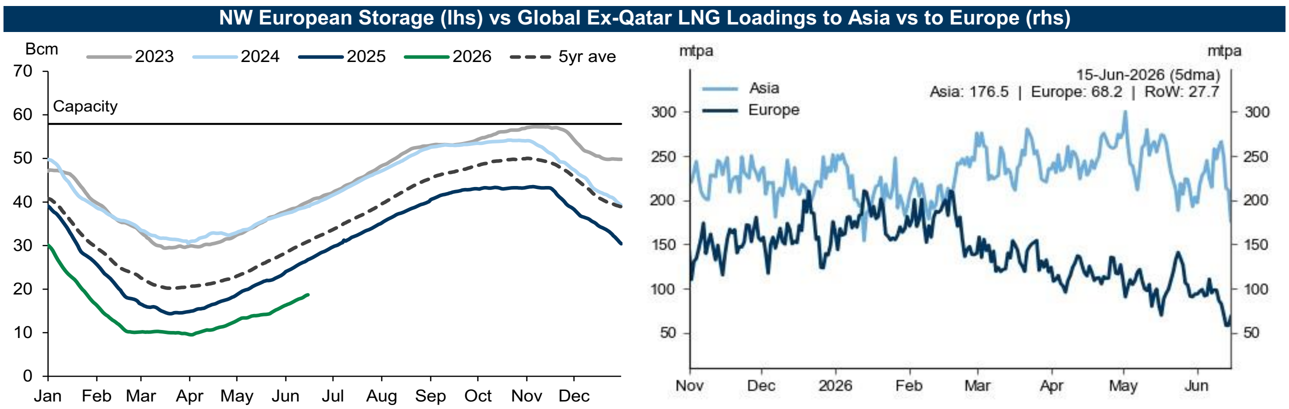
Laura Cyr
+1(212)934-1001 | laura.x.cyr@gs.com
Goldman Sachs & Co. LLC

- We maintain our 2H26/2027 TTF price forecasts largely unchanged at 41/30 EUR/MWh (vs 42/30 EUR/MWh prev., forwards at 42/35) following the announced US-Iran agreement to restore energy flows through the Strait of Hormuz. Although we assume LNG flows will normalize by end-Jul, later than our previous end-Jun expectation, higher ex-Qatar supply and a lower market target for European storage offset the tightening effect from a later reopening.
- First, global LNG supply ex-Qatar has risen more than we expected, partly driven by maintenance avoidance. Second, TTF pricing in recent weeks suggests that the market is targeting a 70%-75% gas storage fill this summer, lower than the 80% we had previously assumed, as LNG supply growth improves winter gas availability. We estimate NW European storage can reach a 74% fill as long as gas prices remain expensive relative to coal, in the low 40 EURs/MWh.
- In 2027, we expect that the combination of a higher end-winter (Mar27) storage vs this year and continued growth in global LNG supply will take NW Europe end-Oct27 storage to a very comfortable 89% full, allowing gas to balance the market while pricing below 35 EUR/MWh coal. Accordingly, we maintain our average 2027 TTF price forecast at 30 EUR/MWh.
- Risks to our 2026/27 gas price view remain skewed to the upside. Should the blocking of Hormuz largely continue, allowing for an only gradual normalization of flows through 2027, we estimate TTF would likely need to move above 100 EUR/MWh this winter (and average above 50 EUR in 2027) to discourage Asia LNG demand. In contrast, even if we see a faster-than-expected ramp of Hormuz flows and start-up of the upcoming Qatari LNG expansion, we estimate only a modest softening of European gas balances likely pressuring 2027 TTF to 27 EUR/MWh.
- Longer term, we maintain our bearish 2028/29 TTF view at 19/16 EUR/MWh (vs forwards at 27/24), with risks skewed to the downside. Recent FIDs of new US LNG export projects signal a larger and longer-lasting LNG oversupply vs what we already embed in our balances. Further, a potential pick up in coal and renewable generation in the aftermath of the Iran conflict might reduce Asia's appetite for natural gas demand in the coming years vs our base case.

US-Iran Agreement Paves Way For European Storage Refill

We had been emphasizing in recent client conversations that European gas markets had been at a crossroads. The loss of Qatari LNG supply was never sustainable for European gas markets owing to the need for gas storage to refill ahead of the winter, especially given the strong rebound in Asia LNG demand since May (Exhibit 1). So, either LNG flows through the Strait of Hormuz would start to normalize to allow European gas storage to refill, or TTF would have to rally to get it done by discouraging Asia LNG demand.

Exhibit 1: Low European gas storage and an ongoing Asia pull for LNG cargoes at the expense of Europe would likely lead to a further rise in TTF without a Hormuz flow normalization



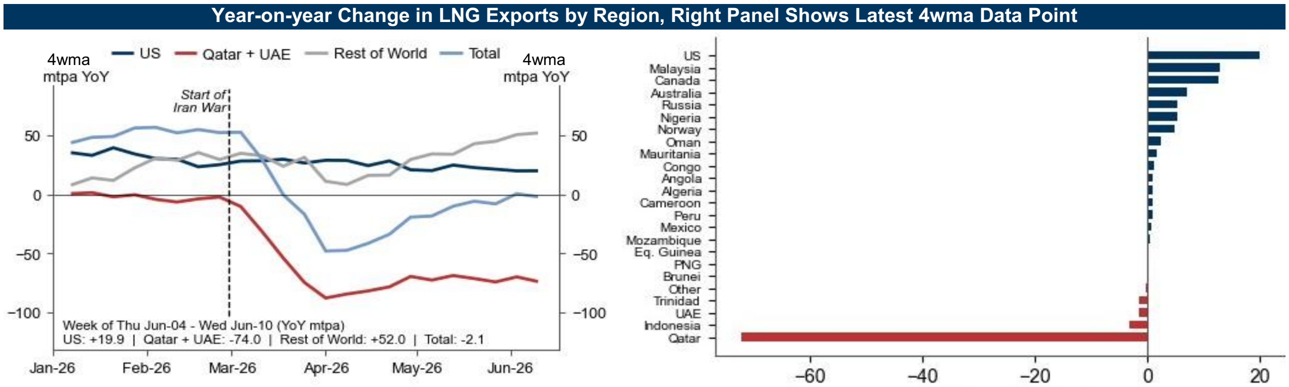
Source: Bloomberg, Kpler, Goldman Sachs Global Investment Research

The recently announced agreement between the US and Iran potentially allows for the former, leading us to keep our 2H26 TTF forecast largely unchanged at 41 EUR/MW (vs 42 EUR/MWh previously, and forwards at 42 EUR/MWh).

Specifically, we now assume LNG flows through the Strait of Hormuz normalize (short of the 13 mtpa long-term damage to Qatari export capacity) by end-Jul, a month later than our previous end-Jun expectation. However, higher ex-Qatar supply and a lower market target for European storage offset the tightening effect from a later reopening.

First, global LNG supply ex-Qatar has risen more than we expected, primarily driven by a combination of apparent maintenance avoidance in places like Oman, Malaysia and Norway, a sharp rebound in feedgas in Nigeria and higher sanctioned LNG supply exports from Russia (Exhibit 2). On net, we raise our Jun26 global ex-Qatar LNG supply forecast by 10 mtpa vs our pre-war expectations to 384 mtpa.

Exhibit 2: Supply growth ex-Qatar has taken global LNG supply back to flat yoy



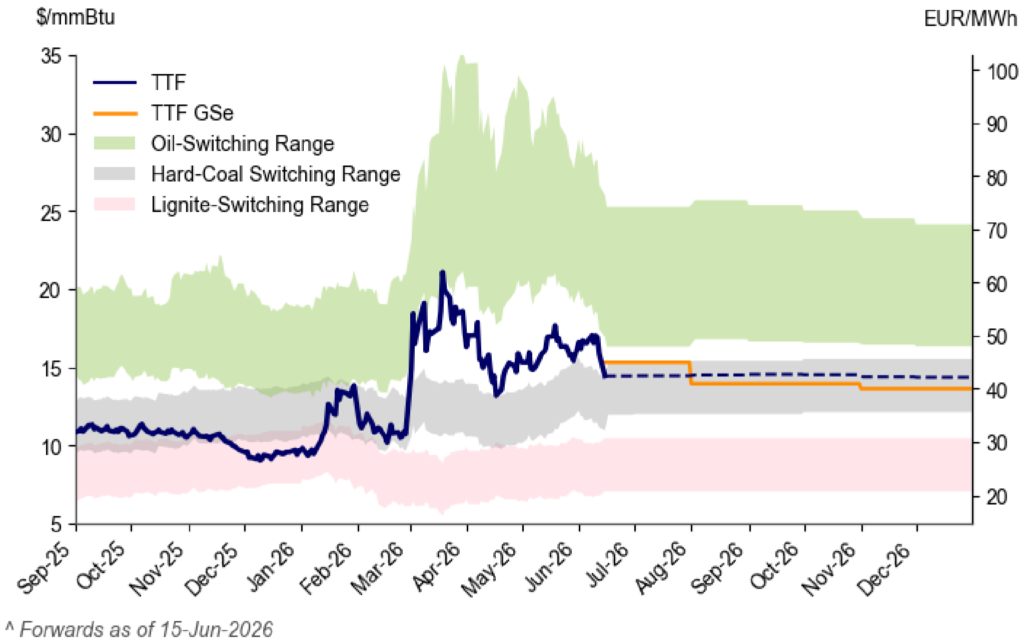
Source: Kpler, Goldman Sachs Global Investment Research

Second, TTF pricing in recent weeks suggests that the market has not been pushing to take storage to 80% full, as we had previously assumed, and appears comfortable taking storage to a lower 70%-75% fill range. We believe this has been driven by growing global LNG supply, which effectively eases gas storage management through winter. Specifically, we estimate that a 70%-75% storage fill ahead of this winter would still allow for an end-winter storage level (assuming 10-year average temperatures) near 30% full, which is high enough to withstand a one-standard deviation colder than average winter, worth 12% of storage capacity in North West Europe.

We estimate NW European storage can reach 74% full by end-Oct26 as long as European gas prices remain expensive relative to coal, in the low 40 EURs/MWh, for the remainder of summer. To be clear, we estimate gas-to-coal switching is maximized when TTF is at least 5 EUR /MWh above coal generation costs. And although coal summer forwards are currently near 40 EUR/MWh, implying forward gas might have to reach 45 EUR to incentivize max fuel switching, we expect coal prices to realize lower as forward risk premium comes off. Prompt coal generation costs have already started to drop, currently at approximately 37 EUR/MWh in gas equivalent terms, which we expect will drop a bit further in the coming months.

Accordingly, we keep our 3Q26/4Q26 TTF price forecasts largely unchanged at 42/40 EUR/MWh (44/40 EUR/MWh prev.) (Exhibit 3). We raise our expected JKM premium to TTF to $2/mmBtu (from $1) in peak summer (Jul-Aug) to reflect the strengthening in Asia pull for LNG cargoes we've observed in recent weeks, leaving our 3Q26/4Q26 JKM price forecast at $16.10/$14.75/mmBtu (vs $15.90/$14.65 prev.).

Exhibit 3: We expect European gas prices to balance the market in the low 40 EURs/MWh for the remainder of summer
Realized and forward TTF prices and fuel switching ranges, and TTF GS fcast



Prices shown in gas-equivalent terms and inclusive of carbon.

Source: ICE, S&P Global Commodity Insights, Goldman Sachs Global Investment Research

Should Hormuz Flows Normalize In Coming Weeks, 2027 Looks Comfortably Supplied...

Going into 2027 we expect that the combination of a higher end-winter (March) storage level vs this year and continued growth in global LNG supply, which we expect up 47 mtpa yoy to 488 mtpa, will take NW Europe end-Oct27 storage to 89% full, a more than comfortable level ahead of the 2027-28 winter. This balance is consistent with TTF prices moving gradually below hard coal levels in the mid-30 EURs/MWh to incentivize higher European gas demand and, in turn, keep gas storage away from congestion. Accordingly, we maintain our average 2027 TTF price forecast at 30 EUR/MWh (below forwards at 35 EUR/MWh). Our 2027 JKM forecast at $11.10/mmBtu is a touch above our previous $10.95 forecast owing to rounding and a minor increase in the JKM premium to TTF assumed for the winter.

Exhibit 4: We maintain our TTF forecasts largely unchanged, as we expect a normalization of Hormuz flows to allow European storage to refill at current price levels
TTF and JKM GS forecasts vs forwards

	GS TTF fcast. (EUR/MWh)	*Previous*	TTF fwrds. (EUR/MWh)	GS TTF fcast. TTF ($/mmBtu)	*Previous*	GS JKM fcast. ($/mmBtu)	*Previous*	JKM fwrds ($/mmBtu)
3Q26	42	*44*	43	14.40	*14.90*	16.10	*15.90*	16.55
4Q26	40	*40*	42	13.75	*13.65*	14.75	*14.65*	15.30
1Q27	38	*38*	41	12.95	*12.95*	13.80	*13.70*	14.40
2Q27	31	*30*	33	10.55	*10.25*	11.30	*11.00*	11.80
3Q27	27	*27*	32	9.20	*9.20*	9.95	*9.95*	11.85
4Q27	25	*25*	32	8.50	*8.50*	9.25	*9.25*	11.95
Bal 2026	41	*42*	42	14.10	*14.25*	15.40	*15.25*	15.90
2027	30	*30*	35	10.30	*10.20*	11.10	*10.95*	12.50
2028	19	*19*	27	6.50	*6.50*	6.90	*6.90*	10.10
2029	16	*16*	24	5.50	*5.50*	5.70	*5.70*	8.95
2030	20	*20*	22	6.75	*6.75*	7.00	*7.00*	8.35
2031	21	*21*	22	7.15	*7.15*	7.45	*7.45*	8.50
2032	21	*21*	22	7.15	*7.15*	7.60	*7.60*	8.75
2033	32	*32*	23	10.90	*10.90*	11.50	*11.50*	9.05
2034	42	*42*	23	14.30	*14.30*	15.40	*15.40*	9.30
2035	42	*42*	24	14.30	*14.30*	15.50	*15.50*	9.60

Forwards as of June 15th, 2026.

Source: ICE, Goldman Sachs Global Investment Research

...Though Hormuz Uncertainty Keeps Risks to 2026/27 TTF Skewed to the Upside

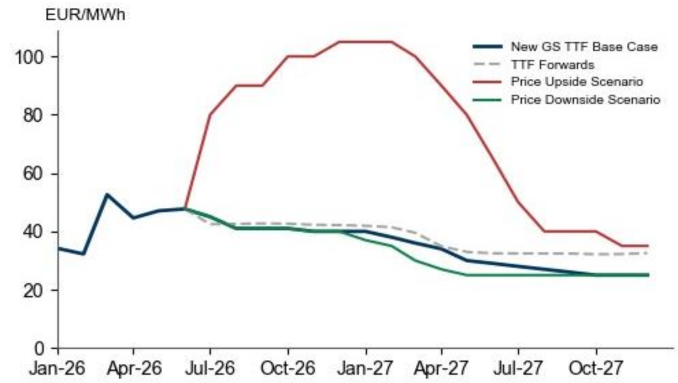
Should the blocking of Hormuz largely continue, if for example the current agreement falls through, allowing for an only gradual normalization of flows through 2027[1], we estimate European storage would likely face significant potential tightness this winter. As a result, we estimate TTF would likely need to move above 100 EUR/MWh this winter (and average above 50 EUR in 2027) to discourage Asia LNG demand.

In contrast, even if we see a faster-than-expected ramp of Hormuz flows (4wks vs 6wks base case) and start-up of the upcoming Qatari LNG export expansion (2Q27 vs our now pushed back base case of 3Q27), we estimate little change to European gas balances beyond taking end-Oct27 storage just above 90% full. Under such a scenario, we would expect 2H26 TTF prices unchanged vs our base case and 2027 average prices only modestly lower, at 27 EUR/MWh, to incentivize incremental gas demand to manage European storage away from congestion.

[1] LNG flows through Hormuz have increased only very gradually since the start of the war and remain well below normal.

Exhibit 5: Risks to our 2026/27 TTF view remain skewed to the upside
EUR/MWh

GS TTF Price Scenarios



	GS TTF Base Case - Normalized Hormuz flows by late Jul26	Price Upside Scenario - Gradual Hormuz normalization ending by Oct27	Price Downside Scenario - Normalized Hormuz flows by mid-Jul26, faster NFE start
3Q26	42	87	42
4Q26	40	102	40
Bal 2026	41	94	41
2027	30	65	27

NFE is the first expansion train currently under construction in Qatar. We assume it comes online in 3Q27 under our baseline and in 2Q27 in our price downside scenario.

Source: ICE, Goldman Sachs Global Investment Research

Longer-term TTF and JKM Risks Remain Skewed to The Downside

We maintain our bearish long-term TTF view, with a 19/16 EUR/MWh forecast for 2028/2029 vs forwards at 27/24 EUR/MWh, and continue to see risks to our view skewed to the downside. Recent FIDs of new US LNG export projects like Commonwealth LNG (9.5 mtpa, 2030 planned start) and phase 1 of Delfin LNG (4.4 mtpa, 2030 planned start) signal a larger and longer-lasting global LNG oversupply vs what we already embed in our balances. Further, a potential pick up in coal and renewable generation in the aftermath of the Iran conflict might reduce Asia's appetite for natural gas demand in the coming years vs our base case.

Team Natural Gas

Samantha Dart
+1(212)357-9428
samantha.dart@gs.com
Goldman Sachs & Co. LLC

Laura Cyr
+1(212)934-1001
laura.x.cyr@gs.com
Goldman Sachs & Co. LLC

Disclosure Appendix

Reg AC

We, Samantha Dart and Laura Cyr, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Samantha Dart Goldman Sachs & Co. LLC, Laura Cyr Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf, and a copy of the annual audit compliance report can be found at this link: https://publishing.gs.com/content/site/india-annual-compliance-report.html. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,